

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2010

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SHOOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

- 4.** Which one of the following components of blood is involved in the clotting after injury?
- A. Plasma
 - B. White blood cells
 - C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D]

14. [A] [B] [C] [D]

24. [A] [B] [C] [D]

34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2010.

1. Which one of the following pests attack seedlings?

A. Stalk borers.

B. Aphids.

C. Cut worms.

D. Weaverbirds.

2. Which one of the following pairs of vessels is correctly matched with the type of blood they carry?

Oxygenated blood

deoxygenated blood

A. Aorta

pulmonary artery.

B. Venacava

pulmonary vein.

C. Pulmonary artery

venacava.

D. Aorta

pulmonary vein.

3. Which one of the following pairs of parasites attack both poultry and rabbit?

A. Mites and lice.

B. Lice and fleas.

C. Mites and fleas.

D. Ticks and mites.

4. Which one following statements about beans seeds is NOT CORRECT?

A. The radicle develops into shoot and plumule into roots.

B. The micropyle allows air and water to enter seed.

C. The hilum is a scar where the seed was attached to the fruit wall.

D. The radicle emerges from the seed before the plumule during germination.

5. The following are some characteristics of a flower:

i) *Small size.*

ii) *Scented.*

iii) *Long feathery stigma.*

iv) *Produce fewer and large sticky pollen grain.*

Which one of the following pairs of characteristics is for a wind pollinated flower?

A. (i) and (iv).

B. (ii) and (iv).

- C. (ii) and (iii).
- D. (i) and (iii).

6. The diagram below represents asset up that can be used to investigate good and poor conductor of electricity.

DIAGRAM

Which one of the following materials when used to connect wires at position K would make the bulb light?

- A. Piece of thread.
 - B. Razor and blade.
 - C. Piece of glass.
 - D. Cello tape.
7. The diagram below represent as set up used by pupils to investigate conditions necessary for seed germination.
Germination occurred in_____.
- A. Q, S and T.
 - B. P and Q.
 - C. R and T.
 - D. R and S.
8. Which one of the following is an adaptation of plants in wet areas?
- A. Freshy leaves.
 - B. Fewer leaves.
 - C. Thick barks.
 - D. Broad leaves.
9. The diagram below represents an arrangement materials in the compost heap.
In the diagram the layer of ash is represented by
- A. K.
 - B. L.
 - C. M.
 - D. N.
10. The diagram below represents a wind sock that was constructed by pupils
Which one of the following is a reason why the instrument could not work?
- A. The sock is made up of strong polythene sheet.

- B. End Q is larger than end P.
 - C. A strong wire was used to make end P.
 - D. A long strong wooden pole was used to suspend the sock.
11. Which one of the following pairs consists only of materials that would not sink in water even when their shape is changed?
- A. Wax and wood
 - B. Wood and aluminum
 - C. Aluminum and glass
 - D. Glass and wax

12. Which one of the following component of air is not correctly matched with its use?

13. The diagram below represents a set up that is used to demonstrate a certain process
The process investigate is

Evaporation

14. Which one of the following pair of diseases are infants immunized against at the 9th month?
- A. Tuberculosis and polio
 - B. Tetanus and whooping cough
 - C. Hepatitis B and diphtheria
 - D. Measles and yellow fever
15. The chart below represents a weather record for five days
Which one of the following statement is true about the weather chart?
- A. Sunny in the morning it was windy in the afternoon
 - B. Cloudy in the morning, it was sunny in the afternoon
 - C. Calm in the morning, it was sunny in the afternoon
 - D. Windy in the morning, it rained in the afternoon
16. Which one of the following statement is true about some levers when in use?
- A. In crowbar, the load is between the effort and fulcrum
 - B. In a crowbar the effort is between the load and fulcrum
 - C. In a wheelbarrow, the fulcrum is between the load and the effort
 - D. In a spade, the effort is between the load and fulcrum
17. Which one of the following statements is not true about HIV/AIDS may spread by
- A. Sharing of unsterilised razor blades
 - B. Coming into contact with other people body fluids
 - C. Shaking hands and hugging infected persons
 - D. Transfusing unscreened blood
18. Which one of the following planets are in the fifth and seventh position from the sun respectively

- A. Mars and Saturn
 - B. Jupiter and Uranus
 - C. Jupiter and Saturn
 - D. Mars and Uranus
19. In the human body water is absorbed in the
- A. Large intestine
 - B. Small intestine
 - C. Stomach
 - D. Rectum
20. Which one of the following pairs consists only substances that are in the same state of matter?
- A. Oil and water vapour
 - B. Wax and glue
 - C. Ice and water
 - D. Air and water vapour
21. Pupils investigate capillarity in different type of soil. Their results were as shown in the diagram below
- From the results the soil samples X, Y and Z are most likely to be
- | X | Y | Z |
|---------|------|------|
| A. Clay | loam | sand |
| B. Sand | clay | loam |
| C. Clay | sand | loam |
| D. Sand | loam | clay |
22. In an experiment to investigate a certain component of soil, water was added to garden soil in the glass container. The component of soil investigated was
- A. Air
 - B. Mineral particles
 - C. Living organism
 - D. Organic matter
23. Tapeworm in livestock are found in the
- A. Large intestine
 - B. Stomach
 - C. Lungs
 - D. Small intestine
24. Which one of the following mixtures can be separated by winnowing?
- A. Flour and husks
 - B. Maize and millet
 - C. Rice and sand
 - D. Millet and husks

25. Which one of the following groups of sources of energy consists of only sources that have chemical energy?
- A. Charcoal, kerosene, food
 - B. Dynamo, drycell, firewood
 - C. Biogas, hydroelectric power generator ,matches
 - D. Geothermal, generator ,cooking gas, car battery
26. Which one of the following groups of sources of energy consists of inclined planes only?
- A. Ladder, spade, crowbar
 - B. Staircase, ladder, a road winding up a hill
 - C. Spade, staircase, ladder
 - D. A road winding up a hill, crow hammer, crowbar
27. Which one of the following is not a method of conserving energy? Using
- A. A wind mill to pump water
 - B. Biogas for cooking
 - C. Solar panel to produce electricity
 - D. A traditional jiko for cooking
28. The diagram below represents a set up that used to investigate a certain aspect of light. The aspect investigated is
- A. Light travels in a straight line
 - B. Refraction
 - C. Reflection
 - D. Passage of light through materials
29. Presence of blood in urine and faces in human being may be a sign of
- A. Typhoid
 - B. Bilhazia
 - C. Malaria
 - D. Cholera
30. Which one of the following pairs of birds have beaks adapted to the same type of feeding
- A. Chicken and duck
 - B. Sunbird and chicken
 - C. Eagle and hawk
 - D. Hawk and duck
31. Which one of the following methods of preserving food is both modern and traditional?
- A. Salting
 - B. Using honey
 - C. Smoking

- D. Drying
32. Which one of the following deficiency diseases is caused by lack of calcium in the diet?
- Anaemia
 - Kwashiorkor
 - Rickets
 - Marasmas
33. Which one of the following statements is not true about commercial feeds animals
- They are given mainly for protection against diseases
 - Some are given as the only feed
 - They mainly to increase production
 - Some are given together with other feed
34. Which one of the following pairs of components of the environment do all animals depend on directly
- Soil and plants
 - Air and water
 - Soil and air
 - Minerals salts and water
35. Which one of the following foods id correctly matched to its group?
- | Food | food group |
|---------------|---------------|
| A. Groundnuts | energy giving |
| B. Eggs | protective |
| C. Carrots | energy giving |
| D. Bananas | body building |
36. Which one of the following animals feeds provides diet of proteins and carbohydrates?
- Clover and Lucerne
 - Sunflower seeds and barley
 - Sunflower seeds and maize grain
 - Lucerne and maize grain
37. The diagram below represents a person pushing a heavy log of wood along a flat surface
Which one of the following changes would make the person push the log more easily?
- Smearing oil on the flat surface
 - Wearing shoes with rough sole
 - Smearing oil on the log of wood
 - Placing rollers on the flat surface
38. The following are some liquids
- Water
 - Kerosene
 - Fresh milk
 - Cooking oil

Which two liquids will mix when put in the container

- A. (i) and (iv)
- B. (ii) and (iii)
- C. (iii) and (iv)
- D. (ii) and (iv)

39. Which one of the following groups consists of materials that are All magnetic?
- A. Staple pins, scissors, metallic bottle top
 - B. Silver coin, knife, marble balls
 - C. Ball bearing, copper coin, razor blade
 - D. Hack saw, steel wool, aluminium
40. Which one of the following practices pollutes soil, air and water?
- A. Use of all farm chemical
 - B. Burning of tyres and plastic
 - C. Dumping of industrial waste
 - D. Use of artificial fertilizers
41. If medicine remains after taking prescribed dose, it is advisable to?
- A. Keep and use it later
 - B. Dispose of the medicine
 - C. Continue taking until it is finished
 - D. Give it to someone with the same sickness
42. Which one of the following is most likely to cause damage to corrugated iron sheets
- A. Aerosol
 - B. Burning farm waste
 - C. sewage
43. Which one of the following diseases cannot be prevented by maintaining proper hygiene
- A. Typhoid
 - B. Bilharzia
 - C. Cholera
 - D. Malaria
44. The beginning of energy transformations in a radio that uses a dry cell is
- A. Electrical
 - B. Magnetic
 - C. Chemical
 - D. Sound
45. The diagram below represents a safety sign when dealing with electricity
- The sign means
- A. Do not insert objects into electric socket
 - B. Do not use appliances with damaged plugs or wire
 - C. Do not plug in many electric appliances
 - D. Do not get near

46. The diagram below represents a set up used to investigate a certain aspect of light.
The aspect being investigated is?
- A. Reflection of light
 - B. Making a rainbow
 - C. Refraction of light
 - D. How light travels
47. Which one of the following is a reason why a glass container is likely to break if hot water is poured into it?
- A. Sudden expansion of the inner side of the glass wall
 - B. Sudden expansion of air in the container
 - C. Sudden contraction of the outer side of the glass wall
 - D. The fact that the heat wall had expand
48. On sea-saw a small boy lift a big boy when
- A. The small boy stands on it
 - B. The two boys interchange position
 - C. The two boys moves closer to the fulcrum
 - D. The two boys move equal distance away from the fulcrum
49. The following are steps followed when investigating the force required to lift a load using fixed pulley but not in their order?
- i) Tie the load with a string
 - ii) Pass the string with the pulley on the support
 - iii) Pull the spring balance and measure
 - iv) Fix the pulley on the support
 - v) Tie the spring balance at the end of spring
- The correct order of steps to be followed is
- A. (ii) (i) (iv) (v) (iii)
 - B. (i) (ii) (iv) (v) (iii)
 - C. (v) (i) (ii) (iv) (iii)
 - D. (iv) (i) (ii) (v) (iii)